

Demonstrating FoodSkin: A Method for Creating Electronic Circuits on Food Surfaces by Using Edible Gold Leaf for Enhancement of Eating Experience

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ABSTRACT

In this demonstration, we introduce FoodSkin, a method for fabricating an electronic circuit on food surfaces by using edible gold leaf. The circuit is easily fabricated using three readily available materials: gold leaf, water, edible paper, and potato starch. Previous methods for enhancing the eating experience, including the presentation of an electrical taste, by making food part of an electronic circuit are difficult to apply to foods with low water content because of their low conductivity. Our method enables the incorporation of dry foods into an electronic circuit by using edible gold leaf to form a circuit on the surface of the food. FoodSkin thus contributes to expanding the design space of the computer-augmented dining experience. We demonstrate the use of FoodSkin for three applications: controlling food taste through electrical stimulation, controlling food temperature, and providing auditory feedback when eating sweets.

CCS CONCEPTS

• Human-centered computing → Human computer interaction (HCI).

KEYWORDS

Gold leaf, eating experience, electrical taste, fabrication.

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1 INTRODUCTION

Gold leaf has long been widely used as a decorative material because of its beautiful luster. Recently, its aesthetics and conductivity have led to it being applied to the surface of various objects to create electronic circuits. In the DuoSkin prototyping and fabrication process [3], gold leaf is adhered to a tattoo sheet to create an electronic circuit for on-body touch interaction. Segawa et al. [11] devised a fabrication technique that uses toner printed by a laser printer as an adhesive to form gold leaf electronic circuits on various substrates. Gold leaf is an edible material, and several methods for forming electrodes and sensors on food products have been reported [9]. However, methods to form arbitrarily shaped patterns using only edible materials have not been established, making it difficult to create complex circuit patterns like those using non-edible materials [3, 11].

In this demonstration, we introduce FoodSkin, a method for fabricating an electronic circuit on food surfaces by using edible gold leaf. Fabrication is simple as it requires only readily available materials; i.e., gold leaf, water, edible paper, and potato starch. FoodSkin enables the integration of dry foods into electronic circuits. It does this by attaching edible gold leaf on the food surface to form a circuit. Existing methods for combining food with electronic circuits, such as for presenting an electrical taste, are limited to foods with higher moisture content because dry foods have low conductivity, making it difficult to apply these methods [6]. Our method is thus a promising way to expand the design space of techniques for enhancing the computer-augmented eating experience. Through the use of FoodSkin, we demonstrated applications that enhance the dining experience by presenting an electrical taste or heat to the tongue via gold leaf electrodes. We also developed a device that generates various sounds depending on the resistance of the gold leaf electrodes.

2 RELATED WORK

2.1 Fabrication of Customized Electronics

Researchers have explored various methods for creating personal circuits and sensors, such as by using conductive ink [4] or copper foil tape [10]. A method has been proposed for food applications that enables the creation of circuits on food surfaces by carbonizing the

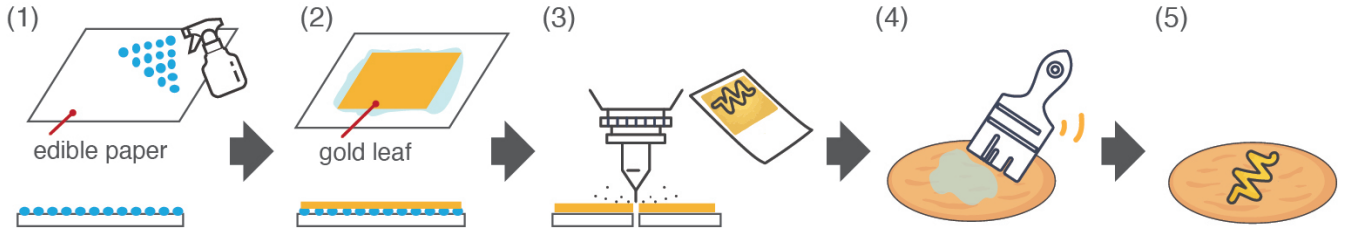


Figure 1: Fabrication of electronic circuit using FoodSkin: (1) moisten edible paper, (2) place gold leaf on paper, (3) cut sheet into desired pattern using laser cutter or scissors, apply adhesive layer to food, and (5) place sheet on surface of food.

material surface through laser irradiation. However, this approach requires applying a phosphate-based flame retardant (non-edible) to the food's surface [12]. The EdiSensor method [9] uses edible gold leaf placed on the food surface as an electrode to detect chewing within the user's mouth. However, the EdiSensor fabrication method is unable to form wiring or sensors in a particular pattern. It simply places gold leaf on food. We have devised an approach for fabricating gold leaf patterns in a desired shape. Templates or parchment paper are used to create electrodes or wires on the food surface to provide interactive elements.

2.2 Enhancing Eating Experience

In the human-computer interface community, several methods have been proposed for controlling food taste and enhancing taste perception. One of them is galvanic taste stimulation (GTS), also known as electrical taste. Electrical taste may indicate the sensation of taste or touch induced by electrical stimulation, but here it refers to a method for inducing, inhibiting, or augmenting the sense of taste by electrical stimulation of the gustatory nervous system. Example applications include enhancing salty taste by applying cathodal stimulation in the oral cavity [6] and enhancing salty taste by applying anodal stimulation from outside the oral cavity via transcutaneous electrical stimulation and thereby achieving an approximately three times stronger taste [5].

Several studies have used tableware-type devices such as forks, chopsticks [5], and bowls [7] as a means of changing taste quality. Another approach is to use a glove-type device [2] to apply GTS to metal tableware. Furthermore, various methods have been proposed that use different current frequencies and electrode surface temperatures to induce various tastes. Ranasinghe et al. proposed placing an electrode at the interface of the tongue and mandible and applying a current to it [7], which raised the possibility of stimulating the taste receptor nerve system on the tongue to induce various tastes. If various tastes can be induced by electrical stimulation, it may be possible not only to enhance the taste of food but also to change the taste at will by applying electrical stimulation simultaneously with food intake. Their method of placing electrodes on the tongue surface can be easily applied to liquid foods, such as soup, and foods containing a lot of liquid because the electrodes cover the tongue surface. In contrast, it is difficult to apply their method to solid foods because their shape does not change. Our FoodSkin method overcomes this limitation.

3 FOODSKIN

The FoodSkin method can fabricate an electronic circuit on solid foods. Gold leaf is a key material due to its conductive and edible properties. By harnessing the capabilities and qualities of gold leaf, we are able to create a functional and safe circuit on food surfaces, opening up diverse possibilities for the integration of technology and culinary arts. Fabrication is easy due to the use of readily available materials (gold leaf sheets, water, edible paper, and potato starch). Previous methods for enhancing the eating experience, including presentation of electrical taste, by including food as part of an electronic circuit are difficult to apply to foods with a low water content due to their low conductivity [1, 6]. In contrast, FoodSkin enables the incorporation of dry foods into an electronic circuit. This is achieved by attaching an edible gold leaf pattern on the food surface to form a circuit.

Fabrication Process.

Since gold leaf is thin, delicate, and difficult to handle [8], we devised a method to strengthen it. We back it with edible paper, which is typically made from rice and other starches. The processing method is illustrated in Fig. 1. It consists of five steps: (1) moisten edible paper slightly using a mist blower, (2) place the gold leaf on the moistened surface of the paper, (3) after the sheet has dried, cut it into the desired pattern by using a laser cutter (we used the beamo laser cutter, Microboards Technology Inc.) or scissors, (4) apply a thin adhesive layer (potato starch dissolved in water and heated to thicken, 2.9–3.3 wt%)¹ to the surface of the food using a paintbrush, (5) place the gold-leaf pattern on the surface of the food so that the edible paper side is in contact with the adhesive layer. Once the adhesive layer is dry, the gold-leaf pattern functions as electrodes or wires. The gold leaf we used had a cost of 0.27 USD/cm², a thickness of 0.12 μm, and a surface resistance of 0.20 Ω/sq.

4 APPLICATIONS

We introduce three applications of FoodSkin that enhance the eating experience. The first is controlling the taste of food through electrical stimulation, the second is controlling the temperature of food, and the third is providing auditory feedback when eating sweets. To retain the original taste of the food, we used gold leaf in the form of a grid or comb to avoid covering a large portion of the food surface.

¹Although edible paper can become viscous when sprayed with water, it has weak adhesive properties and therefore requires the use of an adhesive layer.

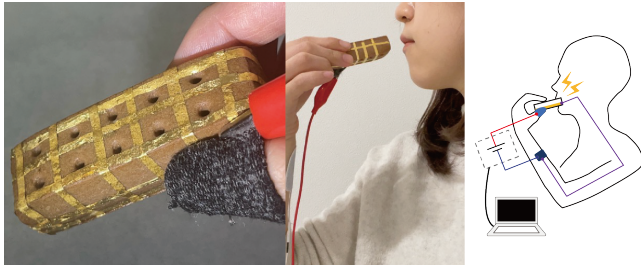


Figure 2: Controlling food taste by applying electrical stimulus to food via edible gold leaf electrode on food surface. Prototype device was constructed using a constant-current circuit and a finger-cot electrode woven with conductive thread.

4.1 Controlling Food Taste through Electrical Stimulation

The first application (shown in Fig. 2) is controlling the taste of a food by applying an electrical stimulus to the food through an electrode made of edible gold leaf attached to the food surface. It has been difficult to control the taste of solid foods with low water content and low electrical conductivity by applying electrical stimulation directly to the tongue or through the food. We addressed this limitation by forming a circuit with gold leaf on the surface of the food. We constructed a prototype device composed of a constant-current circuit and a finger-cot electrode woven with conductive thread [2]. The finger cot has an electrode attached to its surface that contacts the food. The output from the constant-current circuit is transmitted into the oral cavity via the gold leaf.

There are two types of taste stimulation methods for controlling the taste of food: bipolar (both anode and cathode electrodes are applied to the mouth) and unipolar (only one of the two electrodes is applied to the mouth). In the bipolar type, the electrodes on the thumb and index finger each touch two independent electrodes formed on the food. In the unipolar type, electrodes are placed on the thumb and skin (e.g., wrist or neck) and the thumb electrode touches an electrode formed on the food.

When electrodes are placed directly on the tongue, the GTS threshold is widely distributed, from 1–400 μA , and the sensory intensity generally follows the Weber-Fechner law. When the electrodes are placed on the body, the inter-electrode resistance varies greatly depending on the humidity of the skin and electrodes, the size of the electrodes, the position of the electrodes, and the distance between the electrodes. Therefore, we use a constant-current circuit in which a constant current is generated by an operational amplifier and a MOSFET (metal–oxide–semiconductor field-effect transistor), and the current obtained by duplicating the current with a current mirror circuit is used as the stimulating current.

An H-bridge circuit composed of PhotoMOS relays is used to switch the polarity because the current generated by this circuit is of a single polarity; a constant-current diode is used for overcurrent protection. The current waveform and current value are controlled by a digital-analog converter with a microcontroller.

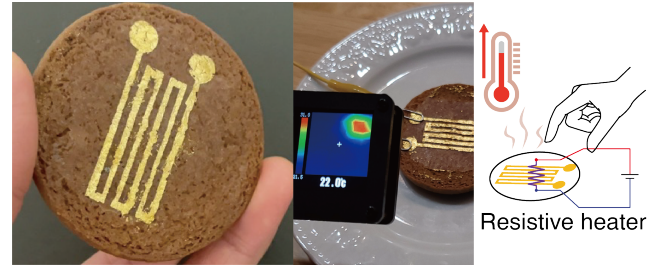


Figure 3: Adjusting temperature of food by using resistive heating element with comb-shaped pattern. As shown in by the thermal camera, it dissipates heat on surface of food.

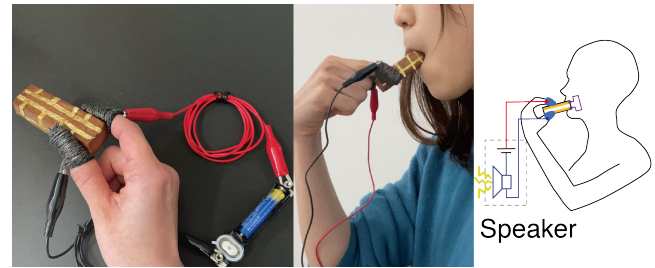


Figure 4: Interactive cookie that emits a sound from the external piezo speaker connected to the grove-type instrument device when consumed.

4.2 Controlling Food Temperature

The second application is adjusting the temperature of food by implementing a resistive heating element with a comb-shaped pattern on the surface of the food. This makes it possible to keep food warm until immediately before ingestion and to change the temperature dynamically during ingestion. Furthermore, we envision that FoodSkin could be used to partially melt temperature-sensitive foods, such as chocolate, by strategically applying heat to specific areas of the food. This would make it possible to alter the texture and shape of the food.

The circuit consists of an electric heating wire created with FoodSkin and a power supply (Fig. 3). We used a conductive pattern with a 1.5 mm width placed at a 2.0 mm interval. The total length is approximately 43.0 mm, and the resistance is approximately 2.6 Ω .

4.3 Providing Auditory Feedback when Eating Sweets

The third application is creation of interactive sweets that emit a sound from an external speaker when consumed. There are two elements: food with a gold leaf circuit and (2) a finger-cot device with an external piezo speaker (Fig. 4). Similar to the setup described in Sec. 4.1, a finger-cot device woven with conductive thread is used, and a grid-patterned electrode is formed on the food surface.

When a user wears the device and eats sweets, various sounds are emitted from the speaker. The sound changes when the microcontroller senses the small current that varies in accordance with the contact between the gold leaf electrode and the tongue or inside

the mouth. This enables the user to enjoy an interaction that dynamically changes the sound while eating. For example, the sound of a crunchy cookie can be enhanced or a character-shaped cookie can emit a sound appropriate for the character when it is eaten. The aim with this application is to motivate children to adopt healthy eating behaviors. The interaction was designed with reference to the EaTheremin fork-type device [1].

5 CONCLUSION

We have demonstrated a method for creating circuits on food surfaces by using edible gold leaf and described potential applications. Our method enables the incorporation of dry food into an electronic circuit, which is difficult with previously proposed methods. We believe that FoodSkin will expand the design space of computer-augmented dining experiences. Future work includes applying FoodSkin to a wider range of applications, such as sensing chewing behavior using capacitive sensing and controlling the pH-induced color change of food by using anthocyanin.

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